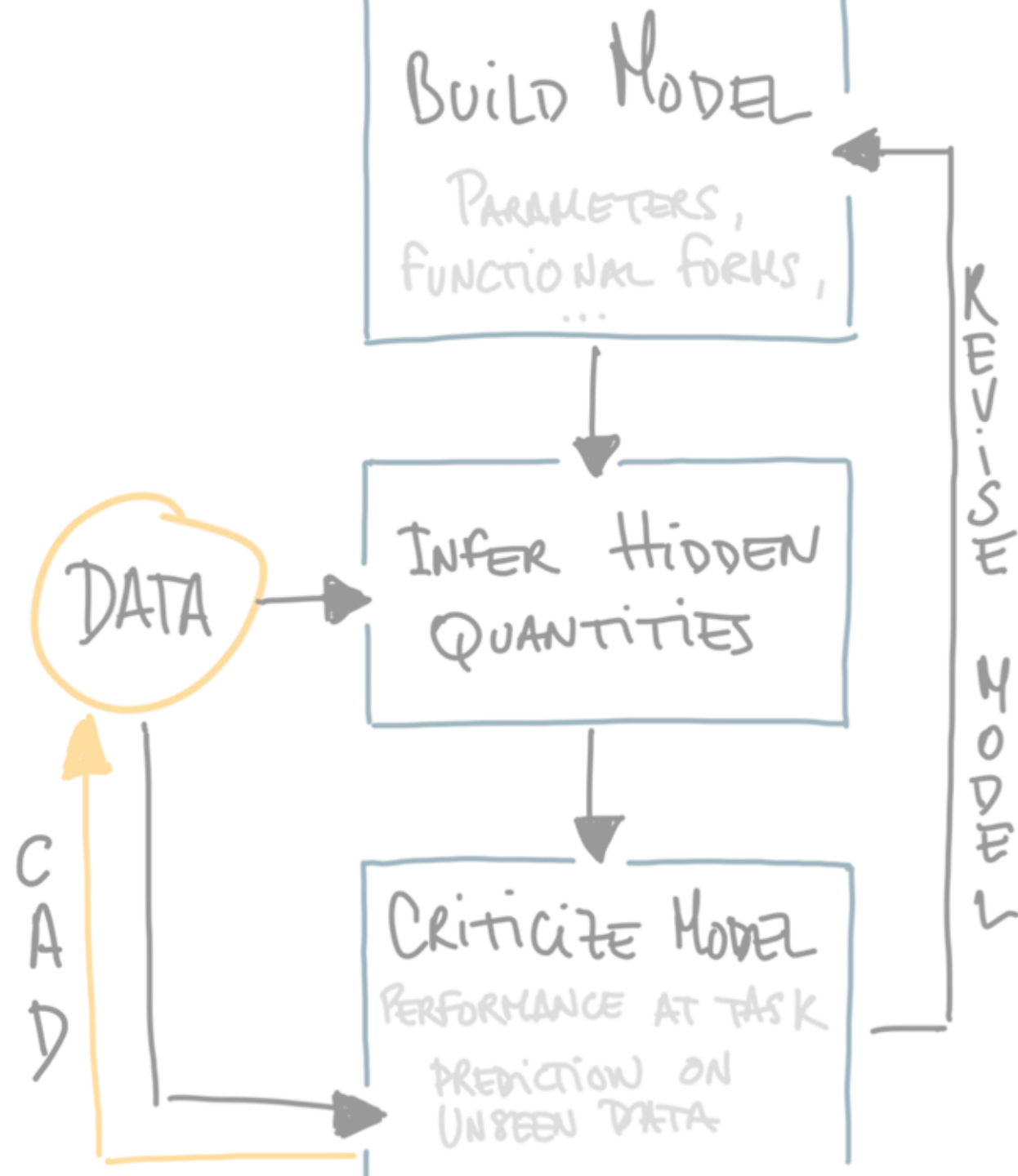
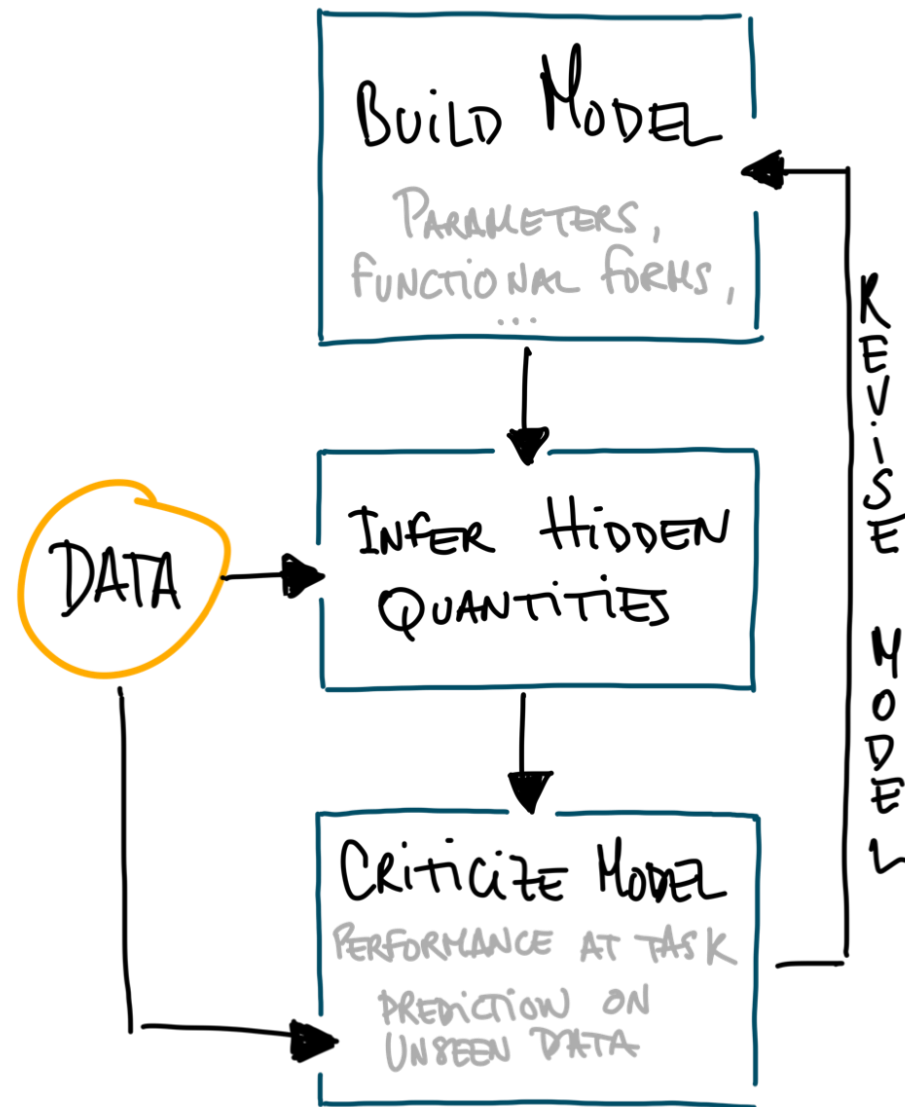


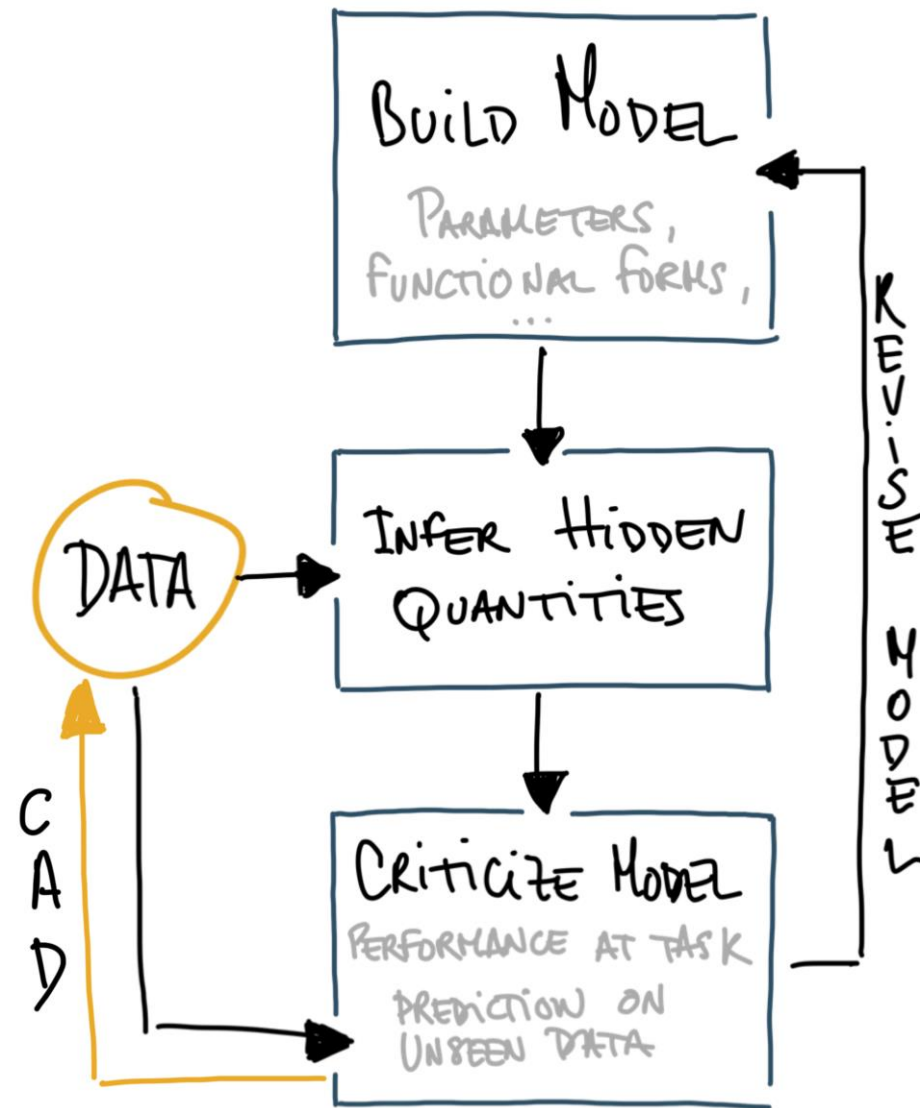
Computer aided experimental design



Back to Box's loop



Back to Box's loop



Bayesian optimal experimental design (BOED)

If we can run experiments iteratively, then we can learn more with less data. BOED is one way to do this – choose the experiment that will reduce your uncertainty the most.

Why does it matter?

- Experiments in biology are expensive, slow, or ethically constrained — you rarely get unlimited tries.
- Optimal design can dramatically reduce the number of experiments needed to pin down a parameter.
- Provides a principled, unified framework for questions that are otherwise answered by intuition or convention.

An example of an adaptive experiment:
The magical number 7

Can you remember this number?

6502803

An example of an adaptive experiment: The magical number 7

Can you remember this number?

6502803

In 1956, George Miller proposed that human short-term memory can hold 7 ± 2 "chunks" of information — one of the most cited and most misunderstood results in cognitive psychology.

Digit span (read someone a sequence of digits, ask them to repeat it back immediately) was the canonical short-term memory task, dating back to 1880, was one of Miller's key examples.

The goal is to infer someone's working memory

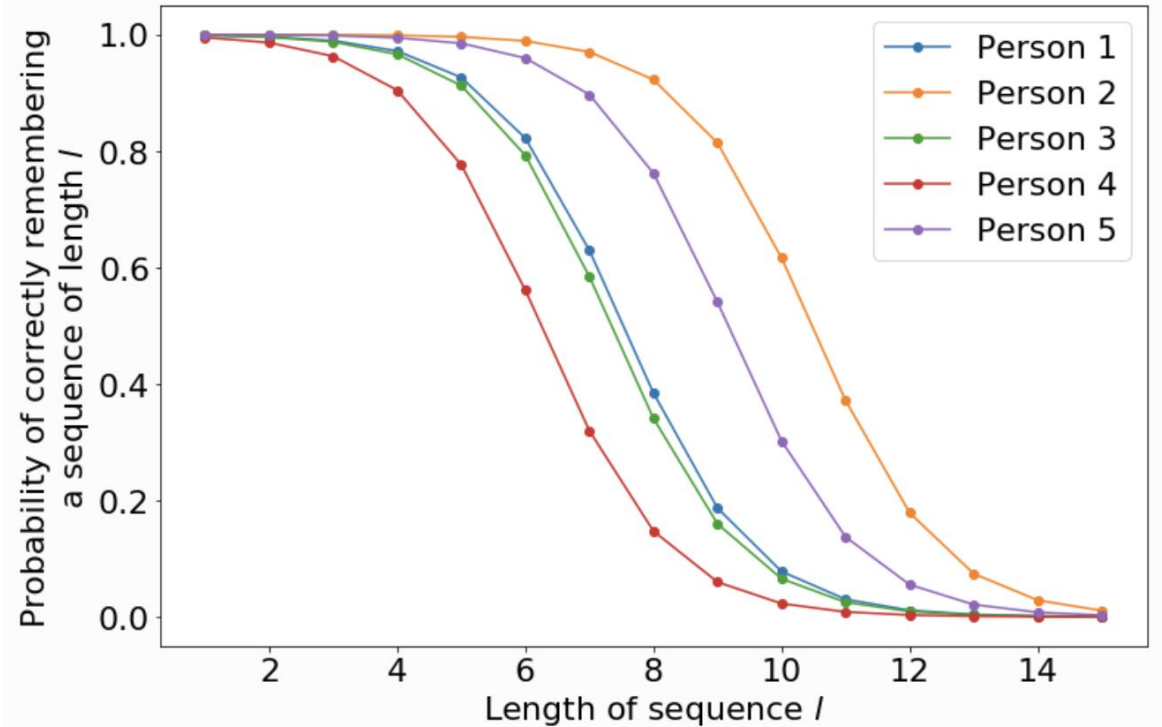
Prior — Miller's 7 ± 2 rule $\theta \sim \mathcal{N}(7, 3^2)$

$$p(l) = \text{sigmoid} [k(\theta - l)]$$

prob of remembering a
sequence of length l

sensitivity

working memory



Inferring someone's working memory with BOED

“Which measurement will teach me the most”

$$\text{EIG}(\underbrace{\epsilon}_{\text{design}}) = \mathbb{E}_{\text{obs} \sim p(\text{obs}|\epsilon)} [\underbrace{KL}_{\text{“distance”}}(\underbrace{p(\text{param}|\text{obs}, \epsilon)}_{?} || p(\text{param}))]$$

Inferring someone's working memory with BOED

“Which measurement will teach me the most”

$$\text{EIG}(\underbrace{\epsilon}_{\text{design}}) = \mathbb{E}_{\text{obs} \sim p(\text{obs}|\epsilon)} [\underbrace{KL}_{\text{“distance”}}(\underbrace{p(\text{param}|\text{obs}, \epsilon)}_{?} || p(\text{param}))]$$

$$\text{EIG}(l) = \mathbb{E}_{\text{obs} \sim p(\text{obs}|l)} [KL(p(\theta|\text{obs}, l) || p(\theta))]$$

↑
Test remembering a
sequence of length l

↑
working memory

Inferring someone's working memory with BOED

"Which measurement will teach me the most"

$$\text{EIG}(l) = \mathbb{E}_{\text{obs} \sim p(\text{obs}|l)} [KL(p(\theta|\text{obs}, l) || p(\theta))]$$

Algorithm

1. Score

$\text{EIG}(l)$ for all $l \in \{1 \dots 14\}$

2. Select

$l^* = \text{argmax}_l \text{EIG}(l)$

3. Observe

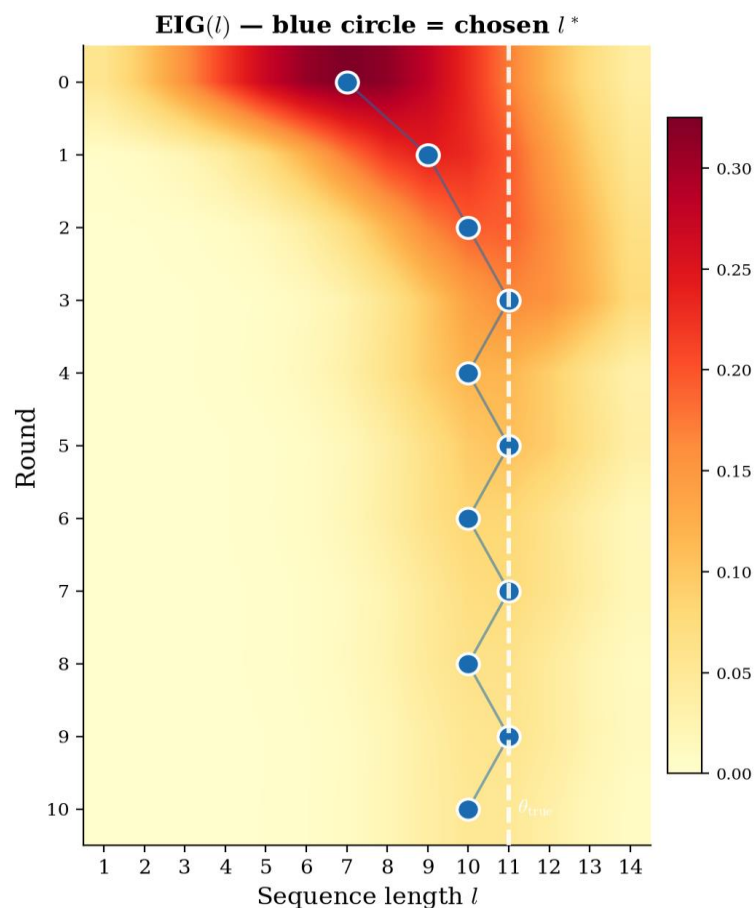
run trial at l^* ; record $y \in \{0, 1\}$

4. Update

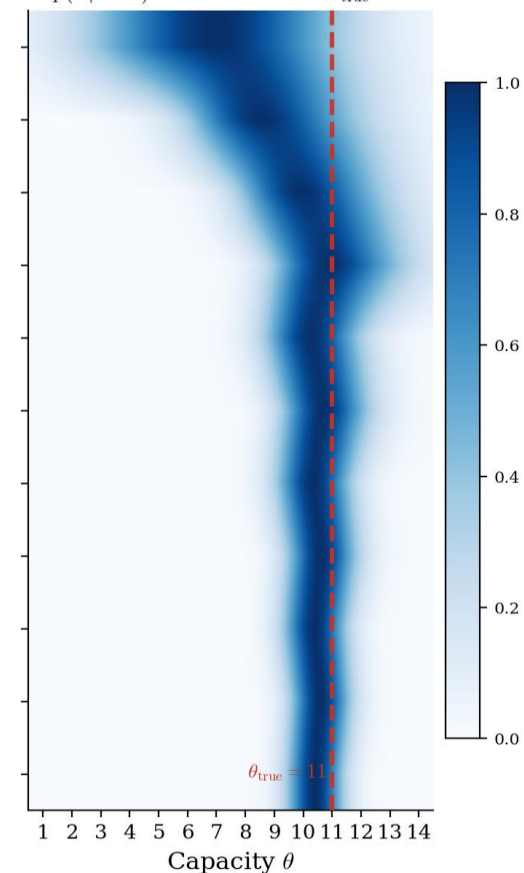
posterior $\leftarrow$ posterior $\times p(y|\theta, l^*)$

5. Repeat

until $\text{SD}(\theta) < \text{target}$



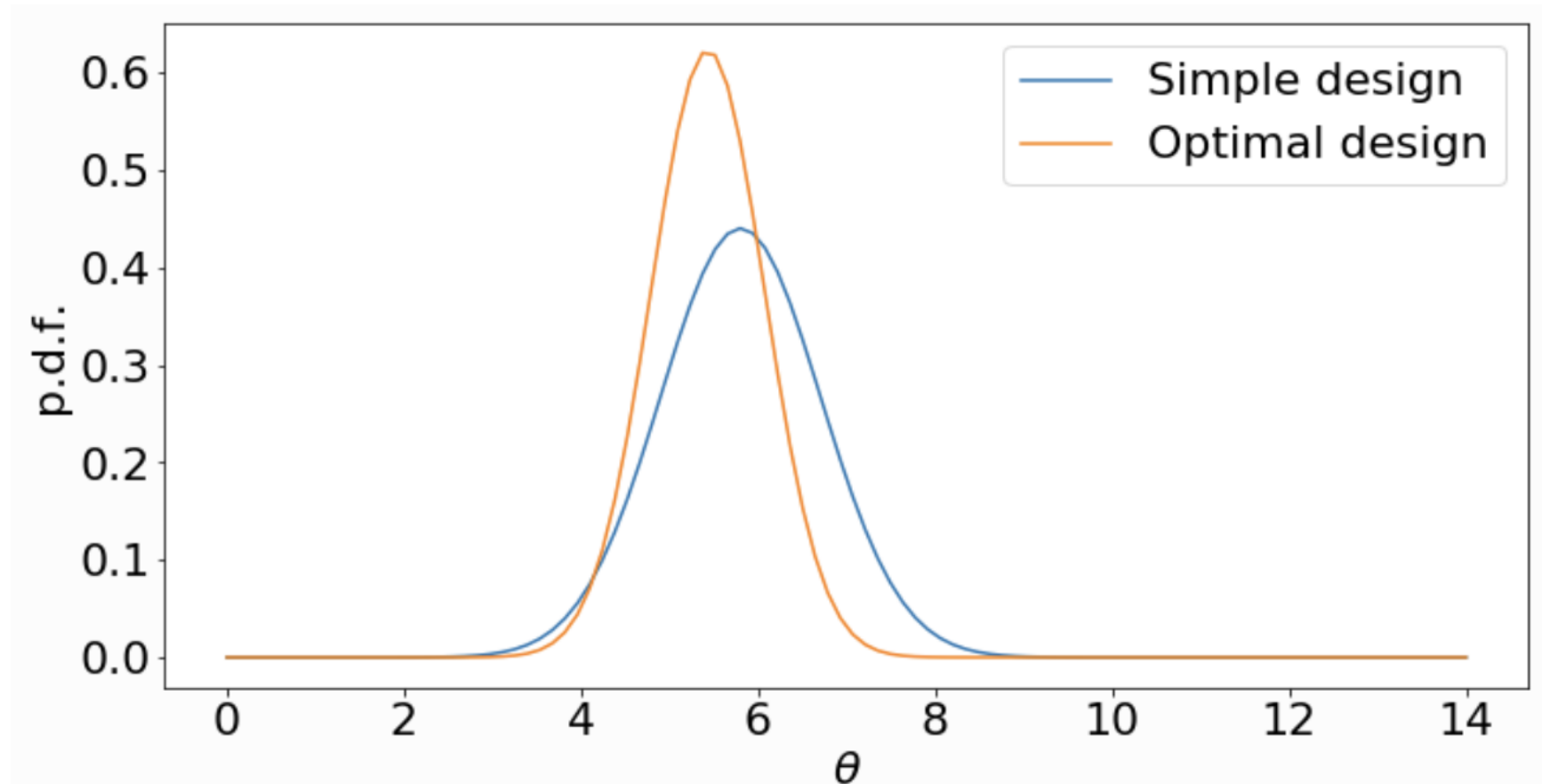
Posterior $p(\theta | \text{data})$ — narrows as θ_{true} is bracketed



Inferring someone's working memory with BOED

“Which measurement will teach me the most”

$$\text{EIG}(l) = \mathbb{E}_{\text{obs} \sim p(\text{obs}|l)} [KL(p(\theta|\text{obs}, l) || p(\theta))]$$



BOED/Active learning/Computer aided design plays an important role in protein design

- Protein design involves searching a vast sequence space – a perfect problem for an integrated approach.
- Deep integration of computation and experiment is now widespread in both academia and industry.

Are more basic biology, or clinical genetics questions amenable to similar strategies?

PROTEIN DESIGN

Robust deep learning–based protein sequence design using ProteinMPNN

J. Dauparas^{1,2}, I. Anishchenko^{1,2}, N. Bennett^{1,2,3}, H. Bai^{1,2,4}, R. J. Ragotte^{1,2}, L. F. Milles^{1,2}, B. I. M. Wicky^{1,2}, A. Courbet^{1,2,4}, R. J. de Haas⁵, N. Bethel^{1,2,4}, P. J. Y. Leung^{1,2,3}, T. F. Huddy^{1,2}, S. Pellock^{1,2}, D. Tischer^{1,2}, F. Chan^{1,2}, B. Koepnick^{1,2}, H. Nguyen^{1,2}, A. Kang^{1,2}, B. Sankaran⁶, A. K. Bera^{1,2}, N. P. King^{1,2}, D. Baker^{1,2,4*}

Article

ProGen2: Exploring the boundaries of protein language models

Erik Nijkamp,^{1,5} Jeffrey A. Ruffolo,^{2,3,5} Eli N. Weinstein,⁴ Nikhil Naik,¹ and Ali Madani^{1,3,6,*}

Low-*N* protein engineering with data-efficient deep learning

Surojit Biswas^{1,2,6}, Grigory Khimulya^{3,6}, Ethan C. Alley^{4,6}, Kevin M. Esvelt⁴ and George M. Church^{1,5}  

Article

De novo design of protein structure and function with RFdiffusion

Article

Illuminating protein space with a programmable generative model

<https://doi.org/10.1038/s41586-023-06728-8>

Received: 20 December 2022

Accepted: 6 October 2023

John B. Ingraham¹, Max Baranov¹, Zak Costello¹, Karl W. Barber¹, Wujie Wang¹, Ahn Vincent Frappier¹, Dana M. Lord¹, Christopher Ng-Thow-Hing¹, Erik R. Van Vlack¹, S Vincent Xue¹, Sarah C. Cowles¹, Alan Leung¹, João V. Rodrigues¹, Claudio L. Morais

Compressed sensing

We are often faced with the situation where **only a subset K of N unknowns is non-zero.**

If we know this a priori, then we don't need N measurements to find the signal.
We only need $O(K \log N)$ random measurements for exact recovery.

Why does it matter?

- Measurements are often expensive, slow, or harmful (MRI scan time, radiation dose, sequencing cost)
- Sparsity is ubiquitous in biology — most genes aren't differentially expressed, most neurons aren't active, most variants don't have large effects

Compressed sensing

Warm-up: Group testing

7 coins, 1 is heavier – how many measurements do you need to make to identify the heavy coin?

Compressed sensing

Warm-up: Group testing

7 coins, 1 is heavier – how many measurements do you need to make to identify the heavy coin?

We can find the coin with 3 measurements

$$\Phi = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Coins to be measured

Measurement

Compressed sensing

Warm-up: Group testing

But what if we have 10000 coins and 14 are heavy?

- How to design an efficient deterministic measurement matrix with theoretical guarantees is an open problem
- Thankfully, we can solve this problem by **using a random matrix!**

$$\Phi = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \rightarrow \Phi_{ij} \sim \text{Bernoulli}(1/2)$$

Number of required measurements is $C \cdot k \log\left(\frac{n}{k}\right)$

Compressed sensing

Compressed sensing generalizes group testing to real-valued and multiple unknowns are non-zero

Group testing (coin problem)

Signal: binary — exactly 1 coin is non-zero

Measurement: binary

Design: each coin gets a unique bit-pattern

Recovery: decode bit-pattern from 3 outcomes

Guarantee: 3 bits $\rightarrow$ 8 patterns $\rightarrow$ 1-in-7 unique

Assumption: exactly 1 non-zero ($K = 1$)

$\rightarrow$

Compressed sensing (general)

Signal: real-valued — K of N are non-zero

Measurement: real-valued — linear combination

Design: random binary / Gaussian A matrix

Recovery: L1 minimisation (Lasso)
or Bayesian sparsity

Guarantee: $M \geq 2K \cdot \log(N/K)$ measurements suffice

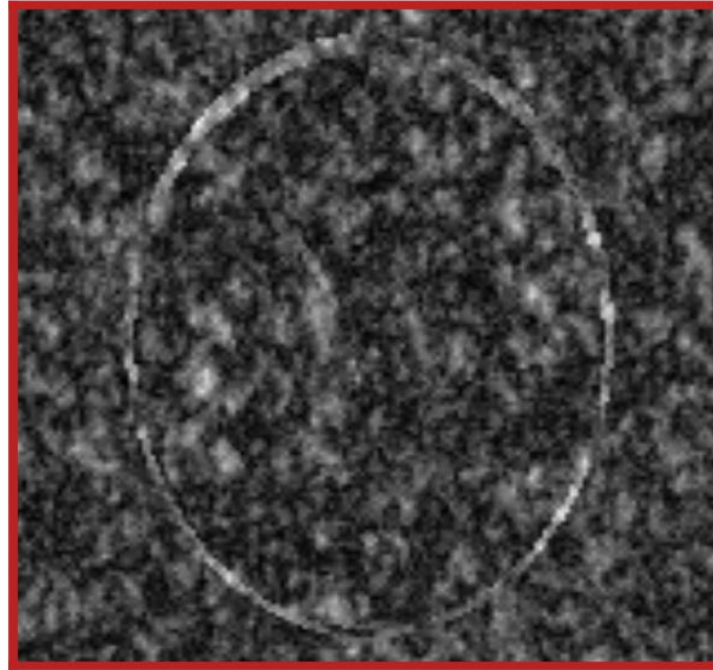
Assumption: $K \ll N$ (sparse signal)

Compressed sensing

Ground truth



Zero-fill (15% k-space)
PSNR = 11 dB



CS recovery (TV prior)
PSNR = 25 dB



What we want

Shepp-Logan phantom — the standard benchmark for MRI reconstruction.

Naive subsampling fails

15% k-space, zero-fill: incoherent aliasing noise throughout the image.

Sparsity prior rescues it

TV minimisation (sparsity prior) preserves edges, removes aliasing. 2-8x faster in clinical MRI.

Only 15% of k-space frequencies are measured. A sparsity prior (TV / L1) recovers near-exact image quality.

Beyond compressed sensing

From binary, to real vectors,...what about function?

Lifting Biomolecular Data Acquisition

Abstract

One strategy to scale up ML-driven science is to increase wet lab experiments' information density. We present a method based on a neural extension of compressed sensing to function space. We measure the activity of multiple different molecules simultaneously, rather than individually. Then, we deconvolute the molecule-activity map during model training. Co-design of wet lab experiments and learning algorithms provably leads to orders-of-magnitude gains in information density. We demonstrate on antibodies and cell therapies.

What is the future of experimental design?

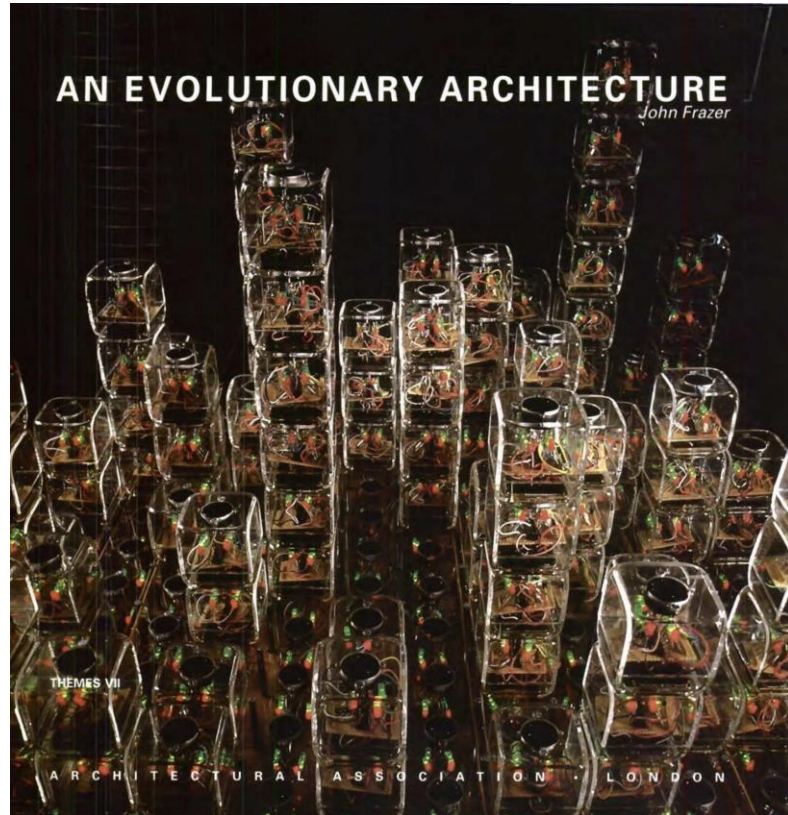
The Road to Biology 2.0 Will Pass Through Black-Box Data

This year marks perhaps the zenith of expectations for AI-based breakthroughs in biology, transforming it into an engineering discipline that is programmable, predictable, and replicable. Drawing insights from AI breakthroughs in perception, natural language, and protein structure prediction, we endeavour to pinpoint the characteristics of biological problems that are most conducive to being solved by AI techniques. Subsequently, we delineate three conceptual generations of bio AI approaches in the biotech industry and contend that the most significant future breakthrough will arise from the transition away from traditional “white-box” data, understandable by humans, to novel high-throughput, low-cost AI-specific “black-box” data modalities developed in tandem with appropriate computational methods.

Michael Bronstein, 2024



What is the future of experimental design?



“The designer moves from the role of creator of a specific form to that of creator of a system or process for generating forms.”

John Frazer, 1995

Session 1: Probabilistic modelling (a Bayesian perspective)

Bayesian models as generative stories

Inductive bias: encoding what you know

Smoothness and sparsity

Notes on inference

Session 2: Latent variable models

Hierarchical models as latent variable models

Variational Autoencoders and ELBO

From VAEs to Diffusion

Session 3: Computer aided experimental design

Bayesian Experimental Design

Compressed sensing